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Flight calculus, twenty terms in plain language

The flight is hypothetical, from Los Angeles to New York. Elapsed time means time since takeoff. No airline record is used.

Calculus connects how fast something is changing with how much has built up.

Name the quantities

A variable is a quantity that can change. On this flight, elapsed time, altitude, fuel remaining, and speed are variables.

A function is a rule that gives one output for each input. Enter an elapsed time into the altitude record and you get one altitude.

An independent variable is the input you choose. A dependent variable is the output you follow. Here elapsed time is the input. Altitude, or fuel remaining, is the output.

A rate of change says how much one quantity changes compared with another. A fuel-burn rate is fuel used per unit of elapsed time.

Slope compares a vertical change on a graph with a horizontal change. On a graph of altitude against elapsed time, slope is vertical speed. It is not how quickly the airplane travels across the ground.

One instant

An instantaneous rate describes one moment. An average rate describes a whole interval. Average ground speed is the path distance traveled, divided by the elapsed time. An instantaneous reading estimates ground speed at one moment.

A limit is the value a quantity approaches as you narrow in on a chosen situation. Shorter time intervals around one moment let you approach its rate. You never divide by a zero-length interval.

A derivative is the exact instantaneous rate in an ideal smooth model. The altitude derivative is zero in level cruise even while the airplane moves quickly across the ground. The height is not changing. The map position is.

Differentiation is the work of finding a derivative. Differentiating a modeled altitude with respect to elapsed time gives a modeled vertical speed.

A tangent line matches a smooth curve's direction at one point. Its slope, on the altitude-time graph, is the modeled vertical speed at that instant. It is not the triangle ratio also called tangent. That ratio is height divided by base, on the ladder page.

Add along the way

An integral adds a continuously varying quantity over an interval. The integral of the fuel-burn rate over elapsed time is the fuel used. It is not the fuel remaining.

Integration is the work of finding an integral. Integrating ground speed over elapsed time gives distance along the path flown. It does not give the straight line between the airports.

On a graph of a positive rate against time, the area under the curve is the accumulated total. Under the fuel-burn graph, that area is fuel used. Area under an altitude-time graph is not a distance traveled.

An initial condition is a known starting value. Start with a known fuel load and subtract the fuel used, and what remains is determined.

A differential equation ties a changing quantity to one or more of its rates. A simple fuel model says the rate of change of fuel remaining is the negative of the burn rate. Negative means the tank is emptying.

Motion, and a choice

Velocity is speed together with a direction. Acceleration is a change in velocity. A turn is an acceleration even when the speed number stays the same, because the direction changed.

A maximum is a highest value in the range you named. A minimum is a lowest value in that range. You might ask for the highest altitude on the modeled trip, or the lowest fuel remaining.

Optimization means choosing the best result you can while obeying the limits you stated. A fuel-saving plan still has to meet safety, weather, and the arrival. Fastest is not the only meaning of best.

A model is a simplified description used for reasoning. The smooth curves are models. Instrument readings are measurements, and they carry uncertainty.

One sentence from the sheet is worth keeping as a summary, not as a slogan pasted over the definitions: the derivative is the rate right now, and the integral is how much has added up.

Key facts

- The flight is invented. Elapsed time is the input. Altitude and fuel are outputs.
- Slope on an altitude-time graph is vertical speed, not ground speed.
- A derivative is an instantaneous rate in a smooth model. Level cruise has altitude derivative zero.
- The integral of burn rate is fuel used. The integral of ground speed is path distance, not the straight-line distance.
- A turn accelerates the airplane even at constant speed, because velocity includes direction.

The sheet is Flight calculus, in plain language (PDF).